REAL: Human-Inspectable Intermediate Representations for AI-Assisted Video Editing

Anonymous authors

Abstract

An incorrect video edit can arise from a mistaken media description, an unsuitable selection or a timing error. An editable project alone does not identify which decision needs repair. REAL organizes AI-assisted editing around four human-inspectable representations: a narrative notebook, media catalogue, authoritative storyboard and editable project. Word anchors, decision ownership and explicit revision commands make choices addressable and record corrections for rebuilding and export. We examine the technical conditions for this interaction contract. Across 30 development and 100 additional synthetic projects, the final frozen implementation reproduces every reference placement under declared assembly semantics. Re-anchoring one overlay in each development project preserves the other event identities and boundaries; corrections also survive rebuilding under two artificial time scales on the same 100-project cohort. In a retrospective comparison, two models supplied with the same plans and format notes also reproduce all 30 projects exactly, locating REAL's contribution in addressable decisions and their correction. Omitting format notes exposes format errors despite close temporal scores. A narrated reconstruction shows a complementary limitation: correct asset selection does not resolve pause and caption conventions. Effects on creators' understanding, effort and agency remain untested. REAL contributes a working inspectable workflow, reproducible conformance evidence and design implications for making multimodal decisions locally correctable.

Keywords

AI-assisted video editing, human-AI collaboration, intermediate representations, multimodal systems, local correction

1. Introduction

A creator assembling a narrated video coordinates what the narration means, what the footage contains, which source interval supports each point, and when images, titles and sound should appear. A misplaced insert may reflect a wrong description, an unsuitable selection or a correct choice translated into the wrong time. Repair becomes difficult when these decisions are collapsed into one generated deliverable.

Consider a violin lesson in which a fingering diagram should appear as the player explains an octave shift. If the diagram arrives late, the creator needs to move it to the relevant phrase while keeping the selected footage, caption style and other placements intact.

Interactive editing systems already connect transcripts to footage [1, 2], describe media in language [3], and expose persistent production artifacts [4]. REAL builds on this direction with an augmentation-oriented stance [5]: the creator remains the author, and intermediate choices should be available for correction. Its systems question is: **How can a creator revise an intermediate editing decision while the workflow preserves that correction through timing and export?**

REAL treats editing as compilation through four readable artifacts. A notebook expresses narrative intent; a catalogue records media evidence and suggested uses; a storyboard owns editorial choices; a compiler translates them into an editable Final Cut Pro project. Models interpret media and requests, while scripts implement binding rules, timing, revisions and validation. A stable word anchor can retain a placement's identity when recorded speech replaces provisional timing; creator ownership marks choices that a rebuild must retain.

Readable plans must survive export, while correctly exported footage may still fail to communicate the idea. We distinguish structural conformance, media selection and creator outcomes.

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We contribute:

1. A working workflow with explicit media evidence, decision ownership and correction rules.
2. Reproducible assembly and local-repair experiments, including frozen 100-project validation and a retained direct-model comparison.
3. A narrated case showing where structural guarantees stop: correct assets still require explicit pause and caption conventions.

These contributions establish technical support for correction. Their effects on editing quality and creator effort remain to be evaluated.

2. Related work

Augmentation and mixed initiative. REAL follows the longstanding goal of amplifying human judgment [6]. Mixed-initiative systems must decide when to act and when to defer [7]; human-centered AI combines automation with human control [8]. Evaluating an augmentation claim ultimately requires people working with the system [9]. We test technical preservation here and leave interaction benefits for a creator study.

Interactive editing. DemoCut structures instructional recordings from lightweight marks [10]. QuickCut connects annotations, transcripts and frame-level selection [1]; B-Script recommends B-roll and insertion positions within a transcript [2]. ExpressEdit supports natural-language and sketch references to temporal, spatial and operational changes [11]. REAL likewise uses language to address editing choices, emphasizing their persistence across preparation, review, retiming and export.

Agent workflows and readable artifacts. LAVE combines footage descriptions with agent assistance and direct manipulation [3]. EditDuet uses Editor and Critic agents to construct and assess sequences [12]. Crayotter exposes retrieval reports, analyses, blueprints and tool outputs in long-form production [4]. These are close precedents: staged agents and inspectable plans are not new contributions here. REAL focuses on who owns a choice, which fields a revision preserves, and independent checks of its translation into a project file.

Its host uses transformer-based models [13], but REAL's representations are authored interfaces rather than learned latent codes. Concept bottlenecks make intermediate concepts available for intervention [14]. Music systems illustrate several ways to keep a person in control: Piano Genie maps an eight-button performance to piano music [15], while Hookpad Aria offers continuations and infilling inside a lead-sheet editor [16]. YuE2 uses symbolic planning before audio generation [17]. These interfaces and plans motivate REAL's editable storyboard; their results do not establish benefits for video editing. REAL applies this principle to media selection and temporal placement without training a new model.

Evaluation boundaries. Message-driven editing benchmarks test selection and sequence agreement [18], whereas QVHighlights supplies annotated video moments [19]. Such tasks differ from perceptual video quality evaluation [20]. Our controlled benchmark isolates identity and timing against a team-authored specification. A small retrieval probe checks a semantic component; neither substitutes for evaluating people finding and repairing consequential mistakes.

3. REAL: representations and correction

REAL supports short narrated explanations assembled from footage, diagrams and voice. Each stage produces a persistent artifact a creator can inspect before translation continues (Figure 1). The design follows documented production failures rather than a formal formative study: expose a decision where its cause can be corrected, separate observations from recommendations, and retain confirmed choices across rebuilds.

A skill combines dialogue instructions, scripts and a handoff format. The host selects the model; no foundation model is built into the four skills. Models interpret media and requests; scripts parse the

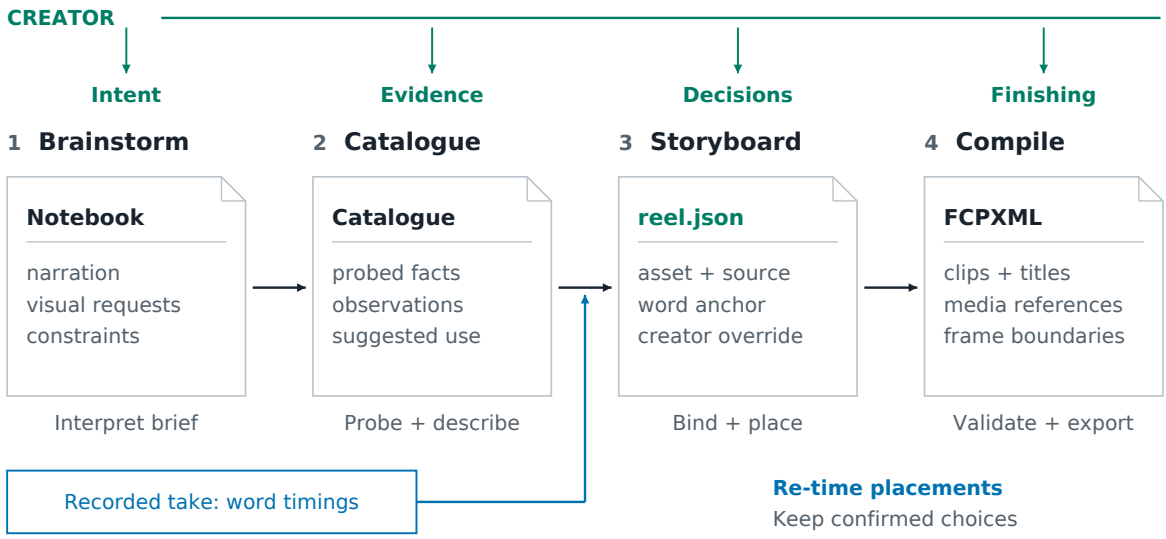


Figure 1: Four persistent artifacts separate narrative intent, media evidence, editorial decisions and finishing. Recorded word times retime placements while retaining confirmed choices.

notebook, probe files, bind assets, derive timing, apply revisions and validate export. The structural tests use supplied descriptions and word times, with no model calls.

3.1. Intent and evidence

A notebook groups typed components into *beats*: narration spans with associated visuals and sounds. Production notes stay outside the narration clock. Requests can specify duration, asset tags and optional status without project-file syntax. This prevents a stage direction from silently becoming spoken time.

Cataloguing records duration, dimensions, frame rate, rotation and audio streams. The production scanner samples 16 evenly spaced frame centres by default; timestamped observations state their inspection coverage. Model descriptions and suggested uses remain distinct from probed facts and can be corrected before binding. Stable asset identifiers preserve references after renaming.

The lexical binder combines explicit tags, declared associations and weighted term overlap, enforcing media-kind compatibility and minimum content evidence. Explicit tags receive priority; type compatibility alone cannot establish a match. Contributing signals and alternatives remain inspectable (Figure 2). Unresolved matches become visible requests or placeholders. An optional grounding stage selects source-event boundaries from ordered observations, records alternatives and refuses unknown or uninspected evidence. Sampling does not establish continuous coverage or precise motion boundaries.

Strongest matches bind first; notebook order and stable asset identifiers break ties. These heuristics make a choice reproducible and challengeable. Their semantic quality is not established by the tagged assembly test.

3.2. Timing and decision ownership

The storyboard is the authoritative edit. Each visual layer records identity, asset, source interval, word anchor, requested duration, style and selector. Audio lanes distinguish narration, source sound, music and effects. The tracks view exposes selected assets and alternatives; edits produce named revision commands rather than an unrecorded replacement of the whole plan.

A *word anchor* identifies a narration token whose time can change. An estimated clock supports planning before recording; a measured clock aligns words to a take. Updating the clock derives new placement times without selecting content again. Creator ownership marks accepted choices that `build --preserve` retains; unrecognized revisions and conflicting choices remain explicit errors.

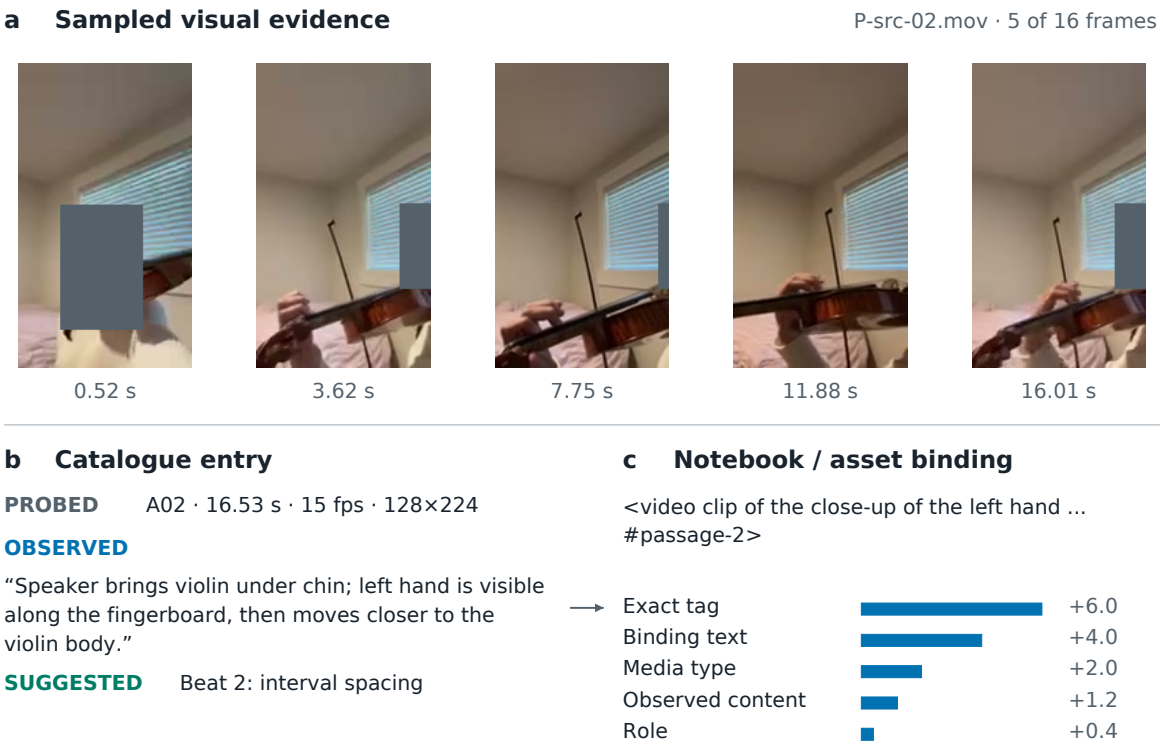


Figure 2: Five of 16 sampled frames from a Paganini lesson proxy and the logged path to an asset choice. Faces are masked for anonymous review. Probed facts, model observation and suggested use remain distinct. The tag and original excerpt window were supplied by the project script; the example does not demonstrate automatic passage discovery.

For word i, the default inline cue is $s_i + 0.4(e_i - s_i) + \delta$, where s_i, e_i are word bounds and δ is the creator's offset. Base shots start at the beat boundary. Nearby cues may start together within a 0.3-second window unless the creator marks one to keep its separate timing; explicit holds extend a beat, while ordinary overlays clip to its available span. The requested duration remains separate so an earlier re-anchor can recover it. A pause policy can hold the base shot until the next beat, subject to source length. These are declared pacing conventions, not learned optimal choices.

Derived times retain precision until export. Absolute boundaries map to frames with Python's ties-to-even rule; duration is the difference of rounded end and start, not a separately rounded length. This avoids a subtle defect: rounding 9.2498 seconds to milliseconds crosses a half-frame boundary at 30 fps.

3.3. Correction and export

In the synthetic walkthrough, `B4: L3 anchor = velvet` moves the same `img07` layer from frames 607–631 to 562–586 at 30 fps (Figure 3). Its requested duration remains 0.8 seconds; the other 15 non-gap placements keep their identities and boundaries, and the new anchor survives rebuilding. The unusual words are test inputs. This demonstrates the implemented correction path, not participant performance.

The compiler checks structure, assets, source bounds and text classification before translating visuals, overlays, titles and audio. Parity checks detect dropped elements; XML checks cover references, frame alignment and timebases. It refuses unresolved decisions rather than inventing editorial replacements. Target conventions still matter: Final Cut positions are percentages of project height, requiring

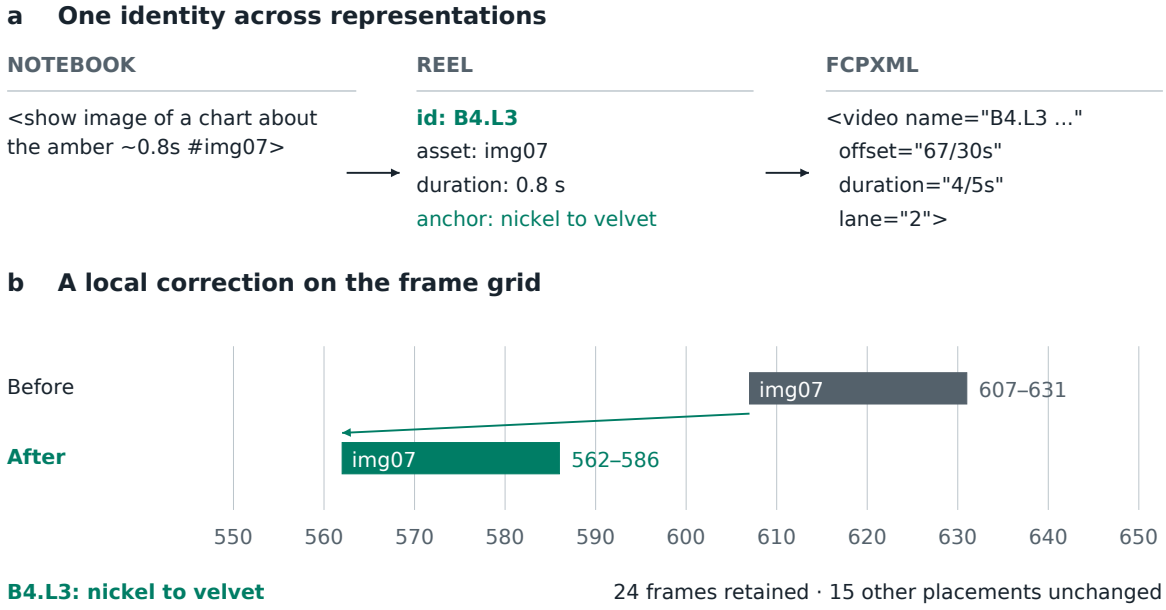


Figure 3: A stable placement across notebook, storyboard and FCPXML. The revision changes its anchor and ownership while retaining the asset. Labelled ranges are absolute; the exported offset is local to the containing clip.

conversion from pixel layout [21]. A development import exposed this error in both exporter and local reader, despite their mutual agreement.

3.4. Recorded-content case

A Paganini lesson complements the synthetic trace with spontaneous speech. A review of the tracks page by the assisting model proposed eight overlay re-anchors to associate captions and stills with named phrases while retaining five base clips. These are proposed editorial choices, not participant repairs. The final export contains the three finished stills and supplied performance opening; its verification script confirms that all 18 tutorial placement and source records shift by exactly 419 frames for the intro. The creator reports that the opening, stills, captions and audio play correctly in Final Cut, including visible Glow. This is a single-project observation, not systematic compatibility evidence.

The case exposed two upstream limits. A project adapter was needed to express narration inside a supplied footage excerpt; downstream source-window support does not by itself solve that notebook authoring gap. PocketSphinx failed on two of five takes with out-of-vocabulary words. On the other three, median onset disagreement with Whisper was 60–90 ms; the largest disagreement was 3.49 seconds on “our” in the fourth take (7.83 versus 4.34 seconds). The project used Whisper word times as its measured clock. Neither ASR source was checked against manual acoustic annotations, so this is a documented fallback and disagreement, not an aligner-accuracy benchmark.

4. Evaluation

We test whether declared choices survive compilation (RQ1), whether one overlay can be corrected locally (RQ2), and which representation errors remain on narrated content (RQ3). These establish technical preconditions for creator intervention. Comprehension, usability and perceived agency require people.

4.1. Controlled assembly and local repair

A seeded generator produces four-to-eight-beat projects with tagged assets, synthetic word times and zero-to-two overlays per beat. Sixteen clips and twelve stills form the media pool. Tags remove semantic ambiguity to isolate timing and compilation. The independent reference imports no pipeline code but shares the team's declared semantics: a beat spans its first to last word, base shots fill it from source frame zero, overlays start 40% into their anchor word and clip to the beat, and absolute boundaries round to the 30 fps grid. It does not exercise the separate hold-after-speech rule.

Initial development uses seeds 1–30; conformance checks use seeds 1001–1100. A changed-clock follow-up reuses the 100-project cohort and exposes two further defects. After fixing them, we rerun assembly, the scripted direct-model comparator and local repairs on one hash-frozen pipeline. These are reused regression cohorts, not an untouched held-out test. Measured word times are supplied before placement. Original-code and estimated-clock runs check benchmark sensitivity and timing dependence; they do not isolate individual implementation changes.

An independently implemented FCPXML reader recovers event identities and boundaries. Greedy one-to-one matching requires the same asset or normalized text and temporal intersection over union (IoU). We report exact-placement F1 (IoU=1), averaged over projects, and the count of all-exact projects. Failed compilation or parsing receives zero and stays in the denominator. Exact matching requires the same identity and both frame boundaries: a one-frame error leaves the prediction and reference unmatched. Percentile 95% intervals use 10,000 paired project-seed bootstrap draws. These metrics exclude appearance and audio mixing.

For each development seed, a fixed rule chooses one overlay and another eligible anchor with enough duration remaining. Its revised bounds are checked against an independent expectation. Collateral change counts non-target event identities whose boundaries appear, disappear or change. It does not measure effects, color, gain or human repair effort.

For the changed-clock check, one eligible overlay per seed is re-anchored, then all supplied word times are scaled by 0.8 or 1.2 in separate copies. Word identities and requested durations stay fixed. We check complete reference placements and saved creator-choice fields after retiming and a rebuild that preserves those choices, and compare two compilations of each retimed story. This tests artificial timing changes, not recordings of new speakers.

4.2. Retained direct-model comparison

Three recorded Claude CLI identifiers—Sonnet 5.5, Opus 5.5 and Fable 5.1—each have 30 retained attempts with and without additional FCPXML format notes. Every prompt supplies the notebook, catalogue, words, revisions and placement rules; models return XML with tools disabled and no repair loop. Thus the task is model-as-compiler, not a workflow without intermediate representations.

We audit all 180 prompts and response/XML pairs. Reference inputs and catalogue asset entries match E10's revised development comparator. Missing or unparseable XML receives zero temporal credit. Parsable XML retains raw scores even when invalid; a secondary validity-gated score assigns zero unless both Apple's 1.13 DTD and the existing semantic validator pass. The reader covers all encountered constructs. Pointwise paired intervals are exploratory, with no multiplicity adjustment or equivalence margin. Regenerated projects and retained final attempts make this a retrospective development comparison; CLI labels do not establish immutable model revisions. Historical title-position instructions also preclude an appearance comparison.

4.3. Narrated reconstruction and semantic probe

A separate 30-second transformer explanation has an agent-prepared reference cut: seven trimmed shots, seven caption images, ElevenLabs narration and a frame-based edit list. REAL receives those labelled assets, notebook and audio; PocketSphinx aligns the supplied words. Saved E05 development modes use blank pauses (A) or hold shots through pauses (B). The reference leads speech by 100 ms and places captions from 400 ms after each cut to 200 ms before the next. Shot and caption errors and

missing picture use the full 900-frame horizon, including any tail after the predicted timeline. Provider word times are a comparison signal, not independently annotated acoustic truth.

A small QVHighlights probe [19] checks semantic selection without reference tags. Five development and five source-disjoint-at-selection queries use one frame per two seconds. REAL describes frames without the query, then grounds the request in those observations; direct vision receives images and the query. Both use the recorded Sonnet 5.5 identifier, but REAL uses two calls versus one. Official evaluator parity was checked across 1,550 sample predictions. Fresh-case uncertainty bootstraps source videos with their queries.

4.4. Reproduction boundaries

The artifact retains frozen implementations, input media, references, predictions and per-case failures. The result table and numerical summaries are generated from saved evidence. The live pipeline can evolve independently of the evaluated snapshots; filenames and a successful current build do not establish version parity. The one-shot supplement includes actual prompts, responses and the executed reader/scorer, with hashes checked before and after replay. An independently extracted replay reproduces its summary exactly.

5. Results

5.1. Assembly and local repair

The frozen measured-clock pipeline compiles and matches every one of 1,269 reference placements in 100/100 validation projects. This establishes conformance to the declared generator. The benchmark exposed five timing defects during development; the original implementation was exact on 0/100 projects.

With estimated word times, 14 of 1,269 placements are exact (about 1%); project-averaged start/end errors are 0.31/0.66 seconds among the 641 placements matched at IoU≥.5 (maximum 2.37 seconds), with 628 unmatched placements excluded from those error averages.

The imported implementation was exact on 28/30 development seeds. Retaining derived precision resolved two one-frame discrepancies without changing the reference or scorer. In the 30 development re-anchor repairs, 30/30 targets have correct revised bounds and 0 non-target identities change boundaries. This supports the tested locality contract, not human repair efficiency.

Under ×0.8 and ×1.2 word-time scales, the correction survives retiming and preserved rebuilding in 100/100 projects each, with saved choices unchanged and repeated compilation byte-identical. The check exposed two one-frame endpoint errors among 200 retimed exports and lost style-ownership labels in 88 projects in the earlier code; both defects were fixed before rerunning the reported snapshot.

5.2. Direct generation from the same plan

Table 1 compares REAL with direct Claude CLI generation on all 30 development projects per condition. Raw temporal F1 retains parsable invalid XML; gated F1 assigns zero unless both the DTD and semantic validator pass. One retained response contains no XML and receives zero in both columns.

With format notes, Opus and Fable match REAL on all 30 projects. The REAL-minus-Sonnet difference is .009 [0.000, .020]. Direct generation can therefore reproduce the supplied plan; equal sample scores do not establish general equivalence. REAL makes validation and rebuilding of saved decisions scripted operations requiring no new model call. Without notes, paired differences are .052 [.016, .093], .036 [.007, .072] and .042 [0.000, .117]. Format failures include still-image element choices, frame-duration declarations and child ordering. These are exploratory development comparisons. All 180 prompt/reference and response-extraction audits pass; a portable frozen replay reproduces the summary byte-for-byte.

Table 1
REAL and direct Claude CLI generation from the same supplied plan (30 development projects per condition). Temporal F1 measures identity and boundaries; format pass requires both validators.

Compiler	Format notes	All exact	Format pass	Temporal F1	Gated F1
REAL	Scripted	30/30	30/30	1.000	1.000
Sonnet 5.5	Supplied	27/30	30/30	0.991	0.991
Opus 5.5	Supplied	30/30	30/30	1.000	1.000
Fable 5.1	Supplied	30/30	30/30	1.000	1.000
Sonnet 5.5	Omitted	24/30	22/30	0.948	0.726
Opus 5.5	Omitted	26/30	26/30	0.964	0.867
Fable 5.1	Omitted	28/30	26/30	0.958	0.867

5.3. Where conformance stops

The narrated reconstruction selects all seven intended shots and caption assets, but timing conventions differ. Blank-pause mode leaves 209 of 900 reference frames without a base picture and ends at frame 875. Holding shots through pauses recovers all seven shot boundaries, while mean caption start/end errors remain 8/6 frames. Source clips were reference-trimmed: this does not demonstrate discovery of those conventions in raw footage. Against provider word onsets, PocketSphinx median absolute error is 51.3 ms (95th percentile 90.9 ms); neither timestamp source is independent acoustic ground truth.

On the five fresh QVHighlights cases, REAL temporal and direct vision each retrieve 3/5 at IoU .5, but mAP is 25.75 versus 32.83. The exploratory source-bootstrap difference is -7.08 percentage points [-24.33, 7.17]. The small, unequal-call probe establishes neither superiority nor equivalence. Detailed predictions and official-evaluator parity are in the artifact.

Frozen source hashes, all-case rows, independent references and reproduction commands accompany the results. No new model inference was used for this analysis. Native import and creator outcomes require their own evidence.

6. Discussion and limitations

6.1. A contract for intervention

The experiments support a limited but useful claim: specified identities and timing can survive compilation and local re-anchoring. Direct models can also compile explicit plans accurately. REAL's proposed value is therefore the addressable decision state and preservation contract, not exclusive ability to produce correct XML. Whether creators understand and benefit from that state is open.

Three design implications follow. **Expose the cause at an editable boundary:** correct a media observation in the catalogue and a placement in the storyboard. **State what a revision preserves:** an anchor change should retain the asset and requested duration, including who chose each saved value; the style-label defect shows that ownership metadata can be lost even when values survive. Replacing an asset may require new source coordinates. **Validate the claim where it matters:** schema, temporal agreement and native appearance need different evidence. The shared coordinate error shows why internal agreement alone is insufficient.

6.2. Limits of the evidence

The team-authored generator covers tagged assets, synthetic words, 30 fps, zero source in-points and constrained overlays. New seeds sample that grammar, not a population of creator projects. The reference shares its specification with the implementation. One-shot outputs are retained development attempts; detailed format notes encode expertise that the compiler already contains. Equal sample scores establish neither general equivalence nor unique need for scripts. Title appearance and native

import remain outside temporal F1. The repair and artificial-clock checks cover re-anchoring on reused fixtures; preservation with new recordings and other revision operations remains open.

The narrated case uses a synthetic voice and reference-trimmed assets. Recovering its shot boundaries depends on the hold convention, while captions remain mismatched. The real Paganini case needed an adapter and alignment fallback; it is development experience, not a participant study. Frame sampling can miss motion or sound, and plausible descriptions can still be wrong. The retrieval probe is tiny, single-model and unequal in call count; public videos may occur in pretraining. We have no full-split QVHighlights result.

6.3. Evaluation with creators

A counterbalanced study could compare correction through tracks and chat using the same footage, model, initial errors and time budget [9]. Tasks would cover incorrect observations, source events and overlay times. Outcomes include localization, repair success, time, collateral changes and perceived control. Readable state could aid repair but impose a learning cost; neither expectation has been tested.

7. Conclusion

REAL makes editorial choices inspectable and locally correctable. The experiments show identity and timing preservation under declared semantics; two direct models also compiled the supplied plans exactly. Narrated reconstruction reveals the need to express pause and caption conventions explicitly. Whether readable intermediate decisions help creators locate and repair errors remains an empirical question.

Declaration on Generative AI

The authors directed the research ideas, experimental design and scientific judgments, with human inspection of videos and Final Cut Pro imports. ChatGPT/Codex, Claude and Gemini supported implementation brainstorming, coding, debugging, and experiment planning, tracking and coordination, alongside manuscript preparation, formatting, paraphrasing, reference checking and citation management. ElevenLabs supplied synthetic narration. The authors retain full responsibility for the work.

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A. Implementation of the skills

The code is public at `github.com/This-Goober/REAL` under `skills/batch-2/skills/`: four skills, 10,508 lines of Python, standard library plus Pillow, ffmpeg and PocketSphinx. This appendix states the representations and rules the results depend on, as pseudo-code where a sentence would be ambiguous.

A.1. Division of labour

Table 2: Who decides what. The model is whichever model the host runs; the structural experiments supply descriptions and word times and call none.

Stage	The model	The scripts	Refuses when
brainstorm	Drafts sequence and mock components with the creator	A words-per-second ballpark, badged as such	Asked for media, ids or times
catalogue	Describes sampled frames; suggests uses	Probes facts; samples 16 frames per clip; groups near-duplicate takes by perceptual hash; assigns stable ids	Sameness is uncertain (asks); unknown facts stay `unresolved`
storyboarding	Interprets loose notes; reviews the tracks page; picks a source event from listed observations	Parse, bind, clock, place, revise, preserve, validate, render the tracks page	No notebook; unknown revision; anchor text and index disagree; unclassified text; a line that would change the clock
compile	Nothing	Adapt, parity, validate, align, assemble voice, report	No reel; a dropped element; a validator error; any request for a time

Handoffs: `NOTEBOOK.md` to `notebook.json` to `bindings.json` and `clock.json` to `reel.json` to `story.json` to the timeline file. Each is validated against a JSON schema on write and on read. The reel is the only editorial handoff; the story is regenerated from it on every run and never edited.

A.2. Vocabulary

Term	Meaning
beat	One narration block with its visuals and sounds; spans first to last word, extended by holds
component	One annotation such as `<show image of two waves>`; typed from its first four words; id is a hash of its text; modifiers `~4s`, `!`, `?`, `#tag`; `|` means simultaneous
word anchor	Token index plus its text; placements store anchors, times are derived; text and index must agree
clock	Word times, estimated from the syllable model or measured by forced alignment; swapping re-derives every placement without re-planning
layer	One visual on one lane: asset, source interval, anchor, requested duration `dur`, occupied span, style, `chosen_by`, runner-ups
ownership	`chosen_by: creator`, set by any revision; preserved by id across rebuilds; a conflict is asked once
stub	A grey placeholder file at the exact size and duration for an unbound component, so the import shows a hole, not missing media

Revision lines, written by the tracks page on a click or typed by the creator, one per instruction:

```
B7: L1 anchor = pressure      B7: L1 source = 12-18      B7: L1 offset = +200
B7: layer L1 = A03            B7: remove L1              B7: confirm L1
B7: stagger L1                B7: add text = "..." anchor 150
B7: <free text>               -> a note on the layer or beat
B7: duration = 6.0 | hold longer | trim 2s   -> refused: the clock is not a placement
```

A.3. Timing

Estimated word duration is speed $(W_0 + K_s$ syllables) with $W_0 = 0.0665$ s, $K_s = 0.1403$ s, plus a pause after punctuation (0.111 s comma, 0.148 s dash, 0.258 s sentence, 0.332 s paragraph, 0.406 s beat). Fitted to 486 recorded words in five sections: 3.11 words per second, worst section error 5.8%. Text durations (0.6 s plus 0.32 s per word, minimum 1.2 s) are conventions, not measurements.

```
derive(plan, words):                # plan is fixed; only word times change
  for beat in plan:
    n0, n1 = start(first word), end(last word)
    beat.t0 = max(0, n0 - longest explicit cold open)
    leads: laid backwards from n0, one cursor per lane; flagged if they no longer fit
    holds: after n1, in written order
    inline p:    p.t0 = n0 if anchored on the first word, else cue(anchor) + offset
    post p:      p.t0 = n1 + offset
    beat.t1 = max(n1, end of holds, end of post layers)
    cued a, b consecutive by t0: if 0 < b.t0 - a.t0 < 0.3 and not b.stagger: b.t0 = a.t0
    base visual: runs to the next base or beat.t1, never under 0.4 s; explicit lengths fixed
    overlay:     t1 = min(t0 + requested, beat.t1), flagged clipped when cut
    every p:     span = t1 - t0; dur keeps the creator's explicit number
  if pause policy = hold: each base runs to 0.1 s before the next beat's first word
cue(i) = s_i + 0.4 (e_i - s_i) for word bounds s_i, e_i

swap(clock, measured):
  s_i, e_i = measured[i] where present (missing words keep estimates, warned)
  derive(plan, words)               # anchors untouched
  report per beat: estimated vs measured span; a coherent same-sign error is a clock fault
```

Measurement runs PocketSphinx forced alignment on the assembled take, with a built-in out-of-vocabulary list for music terms and compounds spelled from their parts; it fails rather than guess when audio and script disagree.

A.4. Frames

```
frames(t) = round_half_even(t * (den / num))   # rate formed first; called once per boundary
duration = frames(t1) - frames(t0)             # never a separately rounded length
```

Snapping happens once, in the adapter; the story carries integer frames and both targets write exact rationals of the frame duration. Rounding derived seconds to milliseconds first moved 9.2498 s across a half frame; evaluating `t * 3000 / 100` turned 8.55 s into 256.50000000000006, rounding to 257 against the reference's 256 (failure 14).

A.5. Binding

```
score(c, a):                          # component c, catalogued asset a
  s = 0
  for tag in c.tags: s += 6 if tag in a.tags else 2.5 if tag in a.text else 0
  for field, cap in (binds_to 4, shows 3, role 2.5, use 2):
    s += min(cap, sum(idf(w) for shared w) / 2.2)        # idf(w) = log((N+1)/(df(w)+0.5))
  if c is text and c.content in a.excerpt: s += 3
  content = s                                           # evidence about content only
  s += kind(c.type, a.kind)   # +2 same; -1.5 / -2.5 image-video; -9 incompatible
  s -= 8 if a.reference, 3 if a.unused, 20 if a.do_not_use, 4 if file missing

bind(c): rank assets by s, ties by id
  text:      own content; an asset attaches only on a verbatim match
  narration: top if s >= 2 and content >= 1, confidence low (confirmed by ear)
  else:      top if s >= 2 and content >= 2.5; high iff s >= 5, margin >= 1.5, kinds agree
             otherwise a stub and a numbered request
```

Type words (image, clip, show, caption) are stop words. Kind alone never binds. Runner-ups and signals are kept for the tracks page.

A.6. Preservation

```
rebuild(notebook, catalogue, old_reel):
```

```
  reel = assemble(bind(notebook, catalogue), derive(plan, words))
  for item in old_reel where chosen_by = creator:
   if item.asset exists: copy asset, anchor, dur, style, style_by, chosen_by into reel[item.id]
    else: ask once
retime(reel, words):
  creator-moved layers keep anchor and length; the rest take derive()'s times
  each beat: t1 = max(narration end, ends of holds and post layers, held pause)
apply(reel, lines):
  refuse a line that changes the clock; apply the rest; mark the item creator's; refit the beat
  report every unparsed line; exit non-zero if anything was refused or unhandled
```

A re-anchor restores `dur` when the beat has room, because `dur` is kept apart from the span the layer last occupied (failure 13). The same inputs compile to the same bytes.

A.7. Compile

```
compile(reel, target):
  validate(reel): schema; anchor text equals the word at its index; a style on every
                  text layer; an asset or a stub on every other layer; audio kinds
  story = adapt(reel): snap boundaries once; resolve media facts; compute no time
  assert layers in = elements out and lanes in = clips out     # parity, or refuse
  file = target.build(story); target.validate(file) reports 0 errors or nothing is delivered
```

The FCPXML validator checks well-formedness and Apple's DTD, reference and id integrity, every time on a whole frame, stills as `video` with zero duration and timed media as `asset-clip`, a contiguous spine, children inside the parent's timebase, in and out points inside the source, and media files present. Children are emitted in the order the DTD's own error text prescribes. Positions are converted from pixels to percent of project height on both axes (failure 16).

A.8. Failure modes the checks catch

Table 4 lists the failure modes the conformance checks are built to catch and the rule that prevents each. Every one produces a valid-looking timeline or is caught only by the compile step's refusal.

Table 4: Failure modes and the rule that prevents each. The assembly benchmark catches 1 to 5 and 13, the narrated reconstruction 7 to 12, the retiming check 14 and 15, the shipped example 6, and an editor import 16.

#	Failure	Kind	Symptom	Rule
1	Base shot cued 40% into its first word	placement	Every beat opens 3 to 4 frames late	A cue on a line's first word lands on the beat boundary
2	A timed still after a running shot taken as the next base	notebook semantics	Shot cut short, up to a third of the beat black	An image interrupting a running shot is an overlay, not a cut
3	Explicit `~1.7s` overwritten by the span the first clock allows	state	Clipped durations stay clipped after re-timing	`dur` is the creator's number; the span is recomputed per clock
4	Instructions say "measure first"; the code can only estimate, then swap	instruction-code drift	Inline versus hold decided on estimated lengths	The clock takes measured times directly; every stated rule has a regression check
5	Re-anchoring a hold leaves the beat stretched	state	Beat overlaps the next; compile refuses	The beat end is recomputed from its contents after every revision
6	Validation splits narration on whitespace, the clock with a regex	interface	A valid notebook fails validation	One shared tokeniser

#	Failure	Kind	Symptom	Rule
7	A visual description ending in "." typed as narration	type inference	Descriptions counted as speech, 2.0 s on a 60 s example	Type from the first four words; a component is never speech
8	The aligner writes `s`/`e`; the clock reads `start`/`end`	interface	The measured path cannot consume the aligner's output	A schema on the measured handoff; the reader accepts both spellings
9	Technical compounds absent from the aligner's dictionary	algorithm	Alignment refuses technical vocabulary	Compounds pronounced from their parts; a built-in out-of-vocabulary list
10	Prose before the first heading becomes an empty beat	representation	Zero-length beat; compile refuses	Leading prose is a note, not a beat
11	Silence between lines is always a black gap	representation	209 of 900 frames black against an editor's cut	The `hold` pause policy
12	A held shot outlasts its clip	representation	A one-frame alignment difference makes compile refuse	The cut moves to where the footage ends, flagged; a clip shorter than its line is refused
13	Re-anchoring a clipped cue keeps the clipped length	state	Requested duration lost after a move	`dur` kept apart from the occupied span
14	Half-frame rounding, 8.55 s to 256.50000000000006 frames	arithmetic	One-frame errors on exact half frames	Rate formed first; one snap per boundary
15	Style ownership relabelled as model-selected on rebuild	state	Values survive, provenance lost	`style_by` on the preservation list
16	Positions written in pixels; Final Cut expects percent of height	target	Overlays off-position while exporter and reader agree	Conversion at the target

Three rules generalise the table. A skill is two artifacts, instructions and scripts, so every rule stated in an instruction file has a regression check. Handoffs drift without types, so every handoff has a schema and the skills share one core module. Where a fixed rule would stand in for something the format cannot say, the notebook gets the word for it, and what remains is a declared convention.

A.9. Tests

A 40-check end-to-end suite runs the scripts on synthetic data, one check per stated rule (35 rules), and 38 unit tests with 48 subtests cover timing precision, layout, geometry, titles, style ownership and grounding. All pass.

A.10. Second target

The sequencing logic never learns an output format: the adapter produces the story, and a target is a module with `build` and `validate`. Two targets exist. FCPXML is the primary. OpenTimelineIO, which DaVinci Resolve and Premiere Pro import natively, is mapped element for element (beats to the first video track, overlay lanes to further tracks, four audio lanes to four tracks). It has no title schema, so text travels as a generator clip with its parameters or, for editors without a title importer, as a transparent PNG drawn at sequence size and calibrated to Final Cut's template (1.78 em per nominal em in a 1080 by 1920 project, block 0.2 em below the glyph centre; 1.73 measured on an export). All 130 benchmark projects read back equivalent through the two independent readers. The Paganini tutorial imported into both editors renders to 2,747 frames in each, all 6 cuts and 22 caption edges on the expected frame, glyph boxes within 1%, median SSIM 0.992 between renders, audio lag 0 ms; a semi-transparent still is 22% darker in Resolve, a renderer property. One tutorial in two editors, not systematic compatibility evidence.